Q1.

a) Consider the following prefix expression! Build an arithmetic expression tree and write the equivalent postfix and infix expressions!

/ / - * a b c + * d e / f ^ g h i

b) Consider the following infix expression! Build an arithmetic expression tree and write the equivalent postfix and prefix expressions!

(a * b) ^ (c / (d * e) - (f / g) ^ (h + i))

c) Consider traversing the ternary tree below using the following `postmid` (RLNM) function. Place the letters forming the string "$M_1UST_1A_1FA_2$ $K_1EM_2A_3L$ $A_4T_2A_5T_3ÜRK_2$" in the nodes of the tree below so as to obtain our great leader's name if you traverse it by the `postmid` function! Assume that the structure TTNodeType is a type formed by accordingly modifying a BTNodeType (the binary tree structure from our lecture slides) to contain a "mid" pointer to point to the middle child of the node.

```
void postmid(TTNodeType *p) {
        if (p!=null){
                postmid(p->right);
                postmid(p->left);
                printkey(p);
                postmid(p->mid);
        }
}
```

Q2.

Consider the C code below. Assuming that *k* characterizes the input size and *m* is a variable independent of the input size (i.e., it does not change with the input size). In terms of *i*, *k* and *m*, what is the execution time of *funcQ1*? (Hint: Consider each simple statement is executed in 1 time unit.)

```
funcQ1(){
   long m, count, t;
   count = 0;
   for (i=2; i<=m; i++){
      t=1;
      while (t<k) {
         t*=i;
         count++;
      }
      for (j=1; j<k; j+=2)
         count++;
   }
   printf("count=%ld",count);
}
```

Q3.

a) Construct a BST from the sequence of keys below!
   12 23 34 82 7 9 5 85 64 68 11 56 60 62
b) Give two more different sequences that give the same BST!
c) What is the best case, worst case, and average case search times of this specific BST!
d) How many depth levels does this BST have?
e) What is the height of the tree?
f) How many different **best** BST topologies (best BST is the BST with the *shortest average* search time for a given set of keys) can you construct from the keys in the above sequence? (Here, consider only the keys! The sequence is not important!)
g) Show one of these best BSTs and give a sequence that provides this BST topology!

Q4.

Consider an n-node best binary tree topology! **Hint:** n: any positive integer!

a) In terms of n, what is the number of the nodes at its deepest level?
b) How many topologies exist that have the same total search time?

Q5.

Consider the recursive function below!

```
void myfunc(int n) {
if (n < 1) {
   return;
}
  for (int j = 0; j < n; j++) printf("%d\n", j);
  for (int j = 0; j < 3; j++) myfunc(n-1);
  for (int j = 0; j < n; j++)  printf("%d\n", j);
  for (int j = 0; j < 4; j++)  myfunc(n-2);
}
```

a) Find, **in terms of n**, the recurrence that represents the execution time *f(n)* for myfunc.
b) Find the **particular solution** for the recurrence with the given initial values *f(0)=0, f(1)=2*.

Q6.

i\j	1	2	3	4	5	6	7	8
1	4	0	12	19	0	39	52	0
2	0	10	15	0	31	42	0	70
3	10	13	0	25	34	0	58	73
4	13	0	21	28	0	48	61	0
5	0	19	24	0	40	51	0	79
6	19	22	0	34	43	0	67	82
7	22	0	30	37	0	57	70	0
8	0	28	33	0	49	60	0	88

Given is the above matrix *m*. In order to represent *m* most efficiently (allocating minimum amount of memory space)) provide ***a partial function formula*** (NOT code!!!) in terms of *i* and *j*.